REQUIREMENTS ENGINEERING

Digital Art Commission & Watermarking Portal

NAME	IBRAHIM KHALEEL
SRN	PES2UG25CS809
DATE	20/08/2026
DOMAIN	Media, Events & Community
ACTORS	Client / Buyer, Digital Artist
PROJECT	Digital Art Commission & Watermarking Portal

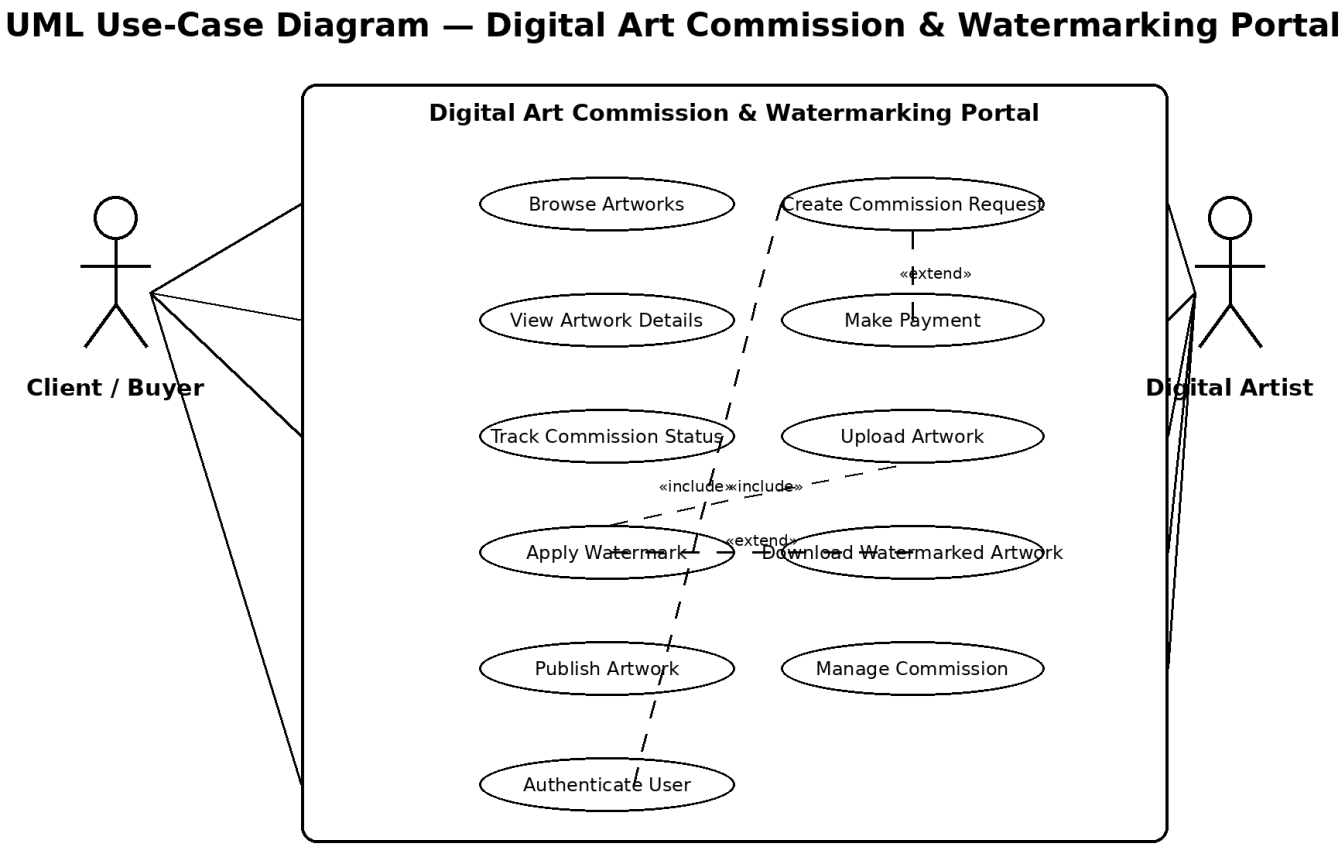
1. FUNCTIONAL & NON-FUNCTIONAL REQUIREMENTS

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow Clients / Buyers to browse published digital artworks and view artwork details.	High	When a Client / Buyer opens the artwork gallery, all currently published artworks shall be displayed with their available details.	Helps buyers discover artists and select suitable digital artwork.
FR-002	Functional	The system shall allow Clients / Buyers to submit digital art commission requests to Digital Artists.	High	When valid commission details are submitted, the system shall create a commission request and make it available to the selected Digital Artist.	Enables buyers to request customized digital artwork.
FR-003	Functional	The system shall allow Digital Artists to accept, reject, and manage commission requests.	High	When an authorized Digital Artist updates a valid commission request, its status shall be updated and visible to the Client / Buyer.	Supports organized commission workflow and communication.
FR-004	Functional	The system shall allow Digital Artists to upload completed artwork and apply a watermark before delivery.	High	When a valid artwork file is uploaded, the system shall apply the configured watermark and make the watermarked version available for delivery.	Protects artists' work from unauthorized reuse before final delivery.

FR-005	Functional	The system shall allow Clients / Buyers to make payment and download the final watermarked artwork.	High	After a successful payment, the system shall confirm the transaction and allow the Client / Buyer to download the permitted artwork.	Provides a complete commission-to-delivery process.
FR-006	Functional	The system shall allow Digital Artists to publish completed digital artworks for browsing.	Medium	When an authorized Digital Artist selects Publish for a valid artwork, the artwork shall appear in the published artwork gallery.	Helps artists showcase their work and attract potential clients.
NFR-001	Nonfunctional	The system shall display artwork gallery, commission, and delivery pages within 3 seconds under normal operating conditions.	High	At least 9 out of 10 page-load tests shall complete within 3 seconds.	Ensures a responsive user experience.
NFR-002	Nonfunctional	The system shall restrict commission-management, artwork-upload, watermarking, and publishing operations to authorized Digital Artists.	High	An unauthorized Client / Buyer attempting to perform restricted operations shall be denied access in 100% of test attempts.	Protects artwork and commission data from unauthorized modification.

2. UML USE-CASE DIAGRAM

The diagram models the two primary actors, their major interactions with the portal, and «include» / «extend» relationships.



3. USE-CASE FLOW SPECIFICATION

Use Case: Submit Digital Art Commission

Use Case ID: UC-01

Primary Actor: Client / Buyer

Preconditions

- The Client / Buyer is authenticated in the Digital Art Commission & Watermarking Portal.
- The artwork gallery is accessible.
- The selected Digital Artist is available for commissions.
- The commission form is available.

Postconditions

- A commission request is successfully created.
- The request is associated with the selected Digital Artist and Client / Buyer.
- The commission status is initialized for further processing.
- The system displays a confirmation to the Client / Buyer.

Main Success Scenario

1. The Client / Buyer opens the artwork gallery.
2. The system displays published digital artworks.
3. The Client / Buyer selects an artwork or Digital Artist.
4. The system displays the relevant artwork and commission information.
5. The Client / Buyer selects the commission option.
6. The system validates the commission details.
7. The Client / Buyer submits the commission request.
8. The system creates the commission request and assigns it to the selected Digital Artist.
9. The system displays a confirmation message indicating that the request was submitted successfully.

Alternate Flow

AF-01: Commission Request Cannot Be Submitted

10. The Client / Buyer submits the commission request.
11. The system detects missing or invalid required information.
12. The system does not create the commission request.
13. The system displays a message identifying the required correction.
14. The Client / Buyer is returned to the commission form.

Use Case: Apply Watermark to Artwork

Use Case ID: UC-02

Primary Actor: Digital Artist

Brief Flow: The Digital Artist uploads completed artwork → the system validates the file → the system applies the configured watermark → the watermarked artwork is stored for protected delivery.